

The transverse motion of solids in rotating cylinders—forms of motion and transition behavior

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Abstract

Mathematical models were developed to predict the transitions between the different forms of transverse motion of free-flowing bed materials in rotating cylinders: sliding, surging, slumping, rolling, cascading, cataracting and centrifuging. Model calculations of the limits between these types of bed motion compare well with measurements of experimental rotating cylinders as well as published results from industrial practice. The motion behavior can be represented on a Bed Behavior Diagram that plots wall friction coefficient and Froude number against the filling degree. From this study, scaling criteria for the bed behavior were found to be the Froude number, filling degree, wall friction coefficient, ratio of particle to cylinder diameter, angle of internal friction, and static and dynamic angles of repose. The transition criteria worked out and the Bed Behavior Diagram provides the user of rotary kilns the possibility to estimate the type of motion of the bed materials used from measured particle characteristics. As a result, the bed behavior can be influenced through selection of operating variables such as rotational speed and filling degree or installation of lifting bars and flights. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Rotary kiln; Transverse bed motion; Bed behavior; Mathematical model; Critical Froude number; Critical wall friction coefficient

1. Introduction

For the design of rotary kiln installations, an estimation of the residence time of solids and of the heat transfer from hot flue gas to the bed is necessary. While the mean solids residence time can already be predicted with sufficient accuracy, suitable fundamentals for calculating the transverse bed motion and the heat transfer from the gas and the rotating wall to the bed have, however, been lacking up to now. Particularly, the heat transport in the interior of the bed has been insufficiently known up to now [1]. Thus, in the heating up zone of the rotary kiln, as a result of the transverse solids motion, a “cold” core in the center of the bed develops, the position and temperature of which cannot be forecast precisely. Above all, this circumstance is due to a lack of knowledge about the internal motion of solids, because the heat transport in the bed occurs mainly through heat absorption to the inclined, gas side bed surface of rolling particles (cascading layer)

and subsequent mixed heat transfer to the colder particle layers in the interior of the bed. Owing to the strong influence of the solids flow, it is to be expected that, just as the segregated core [2], the “cold” core of the temperature distribution is near the vortex center of the agitated bed. A better understanding of the phenomena of transverse solids motion contributes to a more precise calculation of heat and mass transfer in the bed and thus increases safety when designing rotary kilns.

Taking an overview of the transverse solids motion in rotating cylinders as its point of departure, the objective of this paper is to provide simple equations for the calculation of the limits between the different forms of bed motion. The various types of bed motion [1–11,14–21,24]—sliding, surging, slumping, rolling, cascading, cataracting and centrifuging—exhibit significant differences in their mixing behavior, which has an effect on the heat transfer. Knowing the motion behavior of the bed material used as a function of the operating variables can therefore be important for the user of the rotary kiln.

On the basis of simple physical models, transition criteria are derived in the form of critical wall friction coefficients and critical Froude numbers in terms of the filling

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degree. These can be represented conveniently on a Bed Behavior Diagram, a similar form of which was already proposed by Henein et al. [6]. As will be shown, besides Froude number and filling degree, the location of the boundaries between the forms of motion is dependent on the flow properties of the bed to a considerable extent. In order to prove the transition criteria developed, comparisons between model predictions and experimental results have been carried out. For lack of information about the transitions from slipping motion to cascading motion and from rolling to cascading, the author conducted experiments to examine these phenomena. For all other transitions, data researched from the literature have been used to verify the models. All comparisons show a good correspondence between predicted results and data. This study is restricted to unbaffled rotating cylinders and free-flowing monodisperse particle beds.

2. Forms of transverse bed motion

The different types of the transverse bed motion illustrated in Table 1 in summary may be subdivided into three basic forms [7,14]:

- slipping motion
- cascading (tumbling) motion
- cataracting motion.

To delimit the types of bed motion, the ranges for Froude number and filling degree are specified, which,

however, represent orientation values and are dependent on the particular bed material used. The parameter $\mu_{w,c}$ designates the critical wall friction coefficient for the transition from slipping motion to cascading motion. The individual forms of motion are described as follows.

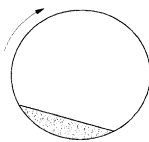
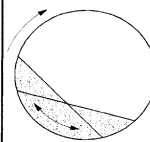
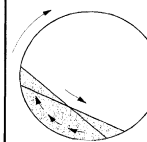
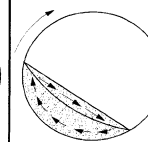
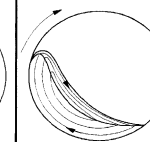
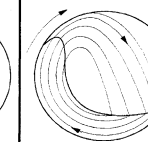
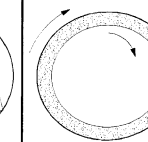
2.1. Slipping motion

Under unfavorable frictional conditions between solid bed and cylinder wall (“smooth tube wall”), slipping motion can occur. There are principally two types of slipping motion [3,7,16]:

- sliding
- surging.

When the cylinder wall is very smooth *sliding* may be observed, which is characterized by a bed constantly sliding from the wall. The tube then rotates under the solid bed, the bed remaining as resting bed under a defined, usually small angle of deflection. According to Rutgers [3], this form of motion (“standing state”) can also occur at higher rotational speeds and filling degrees. With increasing wall friction, sliding turns into *surging*. This type of motion is characterized by periodic alternation between adhesive and kinetic friction of the bed on the wall [3,4]. The solid bed adheres on the rotating wall up to a certain angle of deflection and subsequently slides back en masse on the wall surface.

Table 1
Forms of transverse motion of solids in rotating cylinders

Basic form	Slipping motion		Cascading ("tumbling") motion			Cataracting motion	
Subtype	Sliding	Surging	Slumping	Rolling	Cascading	Cataracting	Centrifuging
Schematic							
Physical process	Slipping		Mixing			Crushing	Centrifuging
Froude number Fr [-]	$0 < Fr < 10^{-4}$		$10^{-5} < Fr < 10^{-3}$	$10^{-4} < Fr < 10^{-2}$	$10^{-3} < Fr < 10^{-1}$	$0.1 < Fr < 1$	$Fr \geq 1$
Filling degree f [-]	$f < 0.1$	$f > 0.1$	$f < 0.1$	$f > 0.1$		$f > 0.2$	
Wall friction coeff. μ_w [-]	$\mu_w < \mu_{w,c}$	$\mu_w \geq \mu_{w,c}$	$\mu_w > \mu_{w,c}$			$\mu_w > \mu_{w,c}$	
Application	no use		Rotary kilns and reactors; rotary dryers and coolers; mixing drums			Ball mills	no use

No particle mixing takes place in slipping motion. As a result, the product quality from rotary kilns decreases. Hence, this state of motion is undesired in practice and must be prevented through the utilization of rough walls or bars attached to the wall. Slipping motion, however, cannot be avoided in every case and may occur at least partially in some production processes.

2.2. Cascading (tumbling) motion

A continuous circulation of the bed, cascading motion, can be observed when there is sufficient wall friction. Dependent on rotational speed and particle size, the following states of motion are possible [2,6,15,20]:

- slumping
- rolling
- cascading.

When the rotational speeds are low, *slumping* of the bed can occur. Through solid body rotation with the rotational speed of the cylinder wall, the solid bed is continuously elevated, being leveled off again and again by successive avalanches at the surface. The slumping frequency is dependent on rotational speed, particle size and cylinder diameter [6,10]. Investigations of Metcalfe et al. [21] revealed that intermixing decreases as the filling degree increases to virtually vanish at $f = 0.5$. As the rotational speed increases a flowing transition to *rolling* takes place. This type of motion is characterized by a uniform, static flow of a particle layer on the surface (cascading layer), while the larger part of the bed (plug flow region) is transported upwards by solid body rotation with the rotational speed of the wall. The bed surface is nearly level and the dynamic angle of repose, which appears only slightly, depends on rotational speed and filling degree [7,8]. This type of motion makes a uniform, good intermixing possible. When particle size distribution is very broad, segregation can appear. As the rotational speed further increases, the bed surface begins to arch and *cascading* sets in. The transition to cascading is also dependent on particle size [8]. The height of the arch of the kidney-shaped bed increases with increasing rotational speed.

The prevailing form of motion in rotary kilns is cascading motion, provided undesired slipping motion can be prevented by creating sufficient wall friction conditions. Then in most cases the rolling bed is preferred, which provides favorable conditions for the heat transfer in high temperature processes and ensures a uniform, high quality of the product even when mass flow rates are large.

2.3. Cataracting motion

As rotational speed increases, the cascading motion is so strongly pronounced that individual particles detach

from the bed and are thrown off into the free space of the cylinder. The release of particles is a characteristic feature of cataracting motion [1–4], which may be subdivided into the following states of motion:

- cataracting
- centrifuging.

Cataracting is characterized by particles from the bed being flung into the gas space. With increasing rotational speed, the number of particles thrown off and the length of the trajectories increase until a uniform trickling veil forms along the diameter. In the case of further increases of the rotational speed, particles on the outer paths $r \approx R$ begin to adhere to the wall and the extreme case of cataracting motion, *centrifuging*, occurs. Theoretically, centrifuging reaches its final stage when the entire solid material is in contact with the cylinder wall as a uniform film. This state is however only achieved, and only approximately as well, at extremely high rotational speeds [12,13]. In a few publications [1,9], cataracting motion is divided into additional types of motion. This form of motion is, however, not relevant for rotary kilns, so that a further subdivision is refrained from.

3. Transition behavior between the forms of transverse bed motion

Numerous publications already exist about the transverse motion of solids in rotating cylinders [1–24], while the transition behavior between the forms of motion has been investigated in only a few papers or handled as a secondary aspect [3–13,24]. This study should contribute to obtaining a complete picture of the behavior of bed motion in rotating cylinders. What is more, existing calculation approaches are further developed or, if necessary, new mathematical models are created. Before doing that, it is expedient to begin with fundamental calculations for the agitated bed in a rotating cylinder.

3.1. Fundamental calculations

Fig. 1 presents the geometric relations in the cross-section of a rotating cylinder in the case of cascading motion. The filling degree as the portion of the cylinder cross-section occupied by the bed is determined by the filling angle as follows

$$f = \frac{1}{\pi} (\varepsilon - \sin \varepsilon \cos \varepsilon). \quad (1)$$

The filling angle ε corresponds to the half bed angle of the circular segment occupied with solids. Assuming a flat bed surface, its distance from the axis of rotation is calculated from $r_0 = R \cos \varepsilon$. Thus, the width (chord) of the solid bed is given by $s = 2R \sin \varepsilon$ and the maximum bed depth at

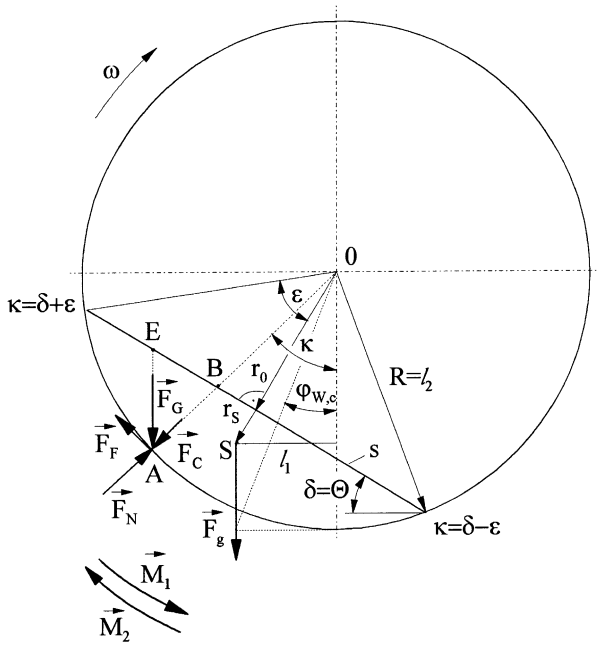


Fig. 1. Schematic of the bed cross-section and moment balance around the cylinder axis.

mid-chord amounts to $h = R - r_0$. The position of the center of gravity of the bed, which in the case of a homogeneous bed is identical with the center of the area of the circular segment surface, is calculated from

$$r_s = \frac{2\sin^3\epsilon}{3\pi f} R. \quad (2)$$

A characteristic criterion for the motion of solids in rotary kilns is the Froude number Fr as the ratio of centrifugal force to gravity. The centrifugal force is related to the inner radius of the cylinder so that this criterion is also named the peripheral Froude number and calculated from

$$Fr = \frac{\omega^2 R}{g}. \quad (3)$$

A solid particle moving in its outermost orbit $r = R$ in the cylinder is centrifuged upon reaching the upper dead point and a critical Froude number of $Fr = 1$. Here the centrifugal and gravitational forces acting on the particle are in equilibrium. The corresponding rotational speed

$$n_c = \frac{30}{\pi} \sqrt{\frac{g}{R}} \approx \frac{42.3}{\sqrt{D}} \quad (4)$$

is designated as “critical rotational speed” and used when specifying the rotational speed of quickly rotating drums such as ball mills [12,22–24]. Hence, the Froude number can also be expressed as $Fr = (n/n_c)^2$. These basic equations are helpful in the development of mathematical models for the transition behavior of the transverse solids motion as follows.

3.2. Transition of slipping motion–cascading motion

The transition behavior between slipping motion and cascading motion was studied in Refs. [4–7] among others. When deriving transition criteria from force balances on a single particle, Reuter [4] considered two cases: motion of the particle on the inclined bed surface as well as on the rotating cylinder wall. The criteria worked out are, however, only partly indicative, since only the consideration of the entire bed and its friction on the wall leads to the targeted result. A limit range for the transition slipping motion–cascading motion of $0 < Fr < 10^{-3}$ was ascertained from experiments [4]. Cross [5] obtained a transition criterion that involves the gravitational and frictional forces on the basis of a moment balance around the axis of rotation. The centrifugal force is disregarded. This criterion relates the filling degree to the flow properties of dynamic angle of repose and bed/wall friction angle. Henein et al. [6] derived a relationship from the force balance around the center of gravity of the bed, which however does not account for the cumulative character of the friction at the entire bed covered cylinder wall. Moreover, according to the assumption $r_s/R \approx 1$, it is only valid for small filling degrees. In Ref. [7], a mathematical model for the cascading motion in rotary kilns was developed, with the help of which simulations were carried out to theoretically ascertain the limits of the range of cascading motion. According to that, the transition from slipping motion to cascading motion with Froude numbers of $Fr \approx 10^{-4}$ is to be expected. Rutgers [3] specifies a relatively high Froude number of $Fr = 10^{-2}$ for this transition. However, as will be shown the filling degree is the main variable influencing the transition slipping motion–cascading motion, not the Froude number. The theoretical approach proposed in Ref. [5] of a moment balance around the cylinder axis is taken up and extended below.

3.2.1. Moment balance around the axis of rotation

Fig. 1 shows a cross-section of a cylinder of unit depth with a solid bed deviating by the angle δ and the moments acting around the cylinder axis. The bed material load causes the counterclockwise moment of the burden M_1 , which tries to turn the cylinder backwards. Frictional forces between the burden surface and the cylinder wall provide the clockwise moment M_2 , which prevents the bed from sliding back. The wall friction is caused by the inherent forces, gravity and centrifugal force. Here in contrast to Cross [5], the centrifugal force is taken into consideration in order to reveal the influence of the Froude number. The moment balance around the cylinder axis results in: $M_2 - M_1 = 0$. If the frictional moment exceeds the moment of the burden

$$M_2 > M_1, \quad (5)$$

the bed moves in rigid body rotation and cascading motion occurs. No distinction is made at this stage as to whether

the motion type is slumping or rolling. Thus, relationship (5) can be used as transition condition to cascading motion. In order to obtain a transition criterion from relationship (5), equations for the acting moments are derived at first.

3.2.1.1. Moment of burden. The gravitational force F_g acts in the center of gravity S of the bed, see Fig. 1. Hence, the counterclockwise moment of the bed material burden about the cylinder axis is calculated from

$$M_1 = F_g l_1 = Mgl_1 \quad (6)$$

with the lever length

$$l_1 = r_s \sin \delta. \quad (7)$$

The mass of bed material in the rotating cylinder amounts to the following from the bulk density and the volume of the bed

$$m = \rho_b f \pi R^2 L. \quad (8)$$

If the filling degree f is constant over the cylinder length, the solid bed can be regarded as a bulk solid body infinitely elongated in the axial direction. The mass of the bed material per 1 m cylinder length thus amounts to

$$M = \rho_b R^2 (\varepsilon - \sin \varepsilon \cos \varepsilon). \quad (9)$$

Hence, the moment of burden is calculated from

$$M_1 = \frac{2}{3} \rho_b g R^3 \sin^3 \varepsilon \sin \delta. \quad (10)$$

3.2.1.2. Moment as a result of wall friction. Frictional forces acting at the entire solid covered part of the cylinder wall result in the clockwise frictional moment

$$M_2 = F_F l_2 \quad (11)$$

with the lever length $l_2 = R$. The complexity of the calculation of the cumulative frictional force F_F arises from the fact that the local forces contributing to wall friction are proportional to the weight of material acting at the wall surface. Actually in a bed of granular solids, in contrast to hydrostatics, the static pressure increases non-linearly with increasing bed depth. This progression may be calculated from Janssen's approximation [26], a widely used formula for the prediction of the vertical stress distribution in bins and hoppers, which obeys an exponential function. In this study the material is assumed to follow hydrostatics with a linear increase of the compressive force over the height. For comparison taking the physical characteristics of cement [26], this approach calculates about 12% higher vertical pressures than Janssen's formula.

According to Coulomb's law of solid body friction, the following applies to the local frictional force

$$F_F = \mu_w F_N, \quad (12)$$

with $\mu_w = \tan \varphi_w$ characterizing the effective frictional coefficient of bed material—wall over the entire solid covered wall. From the radial balance of forces around point A (Fig. 1) the normal force results to

$$F_N = F_G \cos \kappa + F_C. \quad (13)$$

According to the assumption discussed above, the gravity acting on point A is proportional to the length AE; the centrifugal force to the length AB. Thus, the gravitational force is given by

$$F_G = mg = \rho_b g h_1 A, \quad (14)$$

with $A = R d\kappa L$ describing the area of attack and $h_1 = AE$ the height of the bed material over the point A. Using $h_2 = AB$ and $r_{0,x} = OB$ the centrifugal force follows from

$$F_C = m \omega^2 r = \rho_b h_2 A \omega^2 \frac{R + r_{0,x}}{2}. \quad (15)$$

As force per unit surface element of the wall $f_F = F_F/A$, the frictional force is calculated as

$$f_F = \mu_w \rho_b \left(g h_1 \cos \kappa + \omega^2 h_2 \frac{R + r_{0,x}}{2} \right). \quad (16)$$

The differential change of the frictional force over the angle κ results from

$$dF_F = f_F R d\kappa \quad (17)$$

and

$$\begin{aligned} dF_F &= \left(g h_1 \cos \kappa + \omega^2 h_2 \frac{R + r_{0,x}}{2} \right) \mu_w \rho_b R d\kappa \\ &= \varphi(\kappa) d\kappa. \end{aligned} \quad (18)$$

Note that the geometrical terms h_1 , h_2 and $r_{0,x} = R - h_2$ are variables of κ . Through integration of Eq. (18) over the angle κ , thus along the line of contact between the solid and the wall from $\kappa = \delta - \varepsilon$ to $\kappa = \delta + \varepsilon$,

$$F_F = \int_{\delta - \varepsilon}^{\delta + \varepsilon} \varphi(\kappa) d\kappa, \quad (19)$$

the cumulative frictional force amounts to

$$F_F = \mu_w \rho_b g R^2 (1 + Fr) (\varepsilon - \sin \varepsilon \cos \varepsilon). \quad (20)$$

Hence, the moment as a result of wall friction amounts to

$$M_2 = \mu_w \rho_b g R^3 (1 + Fr) (\varepsilon - \sin \varepsilon \cos \varepsilon). \quad (21)$$

3.2.2. Transition criterion

After introducing the Eqs. (10) and (21), the criterion for the transition from slipping motion to cascading motion results from the condition (5)

$$\mu_w > \frac{2 \sin^3 \varepsilon \sin \delta}{3(1 + Fr)(\varepsilon - \sin \varepsilon \cos \varepsilon)}. \quad (22)$$

Cascading motion is reached when the wall friction coefficient μ_w exceeds the right side of the inequality (22). The transition behavior is determined by the variables of wall friction coefficient, angle of deflection of the bed, filling degree and Froude number. The roughness of the refractory wall of a rotary kiln is usually large enough to prevent slipping motion. Slipping can, however, occur in rotary kilns without lining (“smooth steel tube”). For a given bed material, this can be countered by increasing the wall roughness through the installation of lifting bars, by increasing the filling degree or Froude number.

In the case of cascading motion, the center of gravity of the bed deviates from the vertical by the dynamic angle of repose Θ , i.e. $\delta = \Theta$ (Fig. 1). Thus, a formula for the critical wall friction coefficient is obtained through conversion of Eq. (22) to

$$\mu_{w,c} = \frac{2\sin^3\epsilon\sin\Theta}{3\pi f(1+Fr)} \quad (23)$$

As Eq. (23) makes clear for given material properties, the filling degree is the main variable affecting the transition to cascading motion whereas the Froude number, for slow rotation, exerts an admittedly low influence. Neglecting the centrifugal force with $Fr \ll 1$, Eq. (23) corresponds with the criterion derived by Cross [5]. The influence of the filling degree is confirmed in experiments from Korotich [9] on a glass rotating cylinder with quartz sand, which prove that, as filling degree increases, the slipping motion turns into cascading motion. The filling degree at which the transition occurred and the bed material properties are, however, not quoted. The effect of the Froude number has been observed by Lehmborg et al. [15] in experiments on a laboratory rotary drum with a bed of slightly sticky soda particles. The cylinder consisted of an unbaffled steel tube of 0.31 m in diameter. These investigations revealed that, as the rotational speed increases ($Fr > 6.9 \cdot 10^{-4}$), slipping motion turns into cascading motion. However, the more relevant filling degree was not named. The effect of the Froude number is also used in unbaffled ball mills, the rotational speed of which, to compensate for slip, is increased by 5% of the critical in comparison with other mills [27].

For the case of low rotational speeds prevailing in rotary kiln processes (usually $Fr < 10^{-2}$), the progression of Eq. (23) is presented in Fig. 2. The critical wall friction coefficient is depicted in terms of the filling degree and the dynamic angle of repose in the range of $25^\circ < \Theta < 40^\circ$, which is customary in particle technology. As the graph shows, critical conditions for the occurrence of slipping motion are only to be expected at low filling degrees. As the filling degree decreases, the calculated wall friction coefficient clearly increases and can reach the order of magnitude of measured values in rotary cylinders. If the filling degree approaches zero, the wall friction coefficient

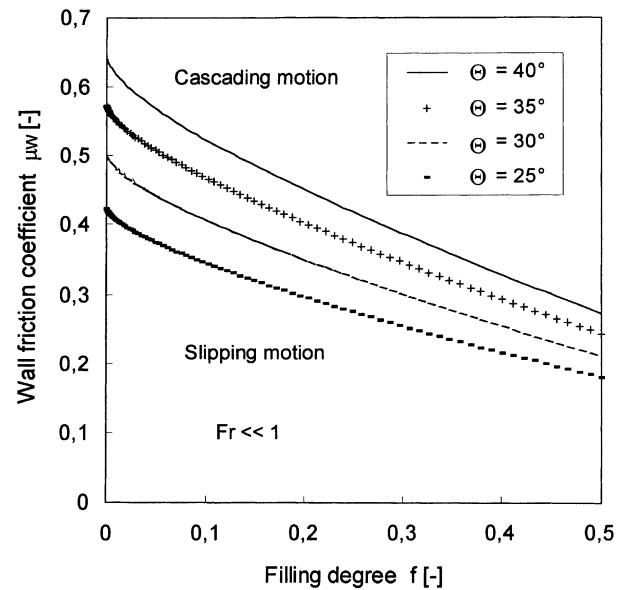


Fig. 2. Transition slipping motion–cascading motion: critical wall friction coefficient in dependence on filling degree and dynamic angle of repose.

reaches its maximum with $\mu_w = \sin\Theta$. For comparison of Eq. (23) with experimental results, see Section 4.

3.3. Transition behavior within the range of cascading motion

3.3.1. Transition of slumping–rolling

Compared with the rolling bed, slumping is characterized by a smaller solids mixing. Hence, it can be significant for the rotary kiln user to know the boundary between the two in order to prevent slumping through the alteration of the variables. Only a few publications [4,6,10,11] exist on the transition behavior of slumping–rolling. Wahlster et al. [10] studied this transition experimentally in a model rotary kiln on different materials that are used in iron ore reduction. It then turned out that the critical rotational speed or Froude number, respectively, is dependent on the particle diameter but to a large extent independent of the filling degree and the bed material being studied. Reuter [4] carried out experiments to ascertain the slumping frequency, i.e. the number of successive avalanches at the bed surface per minute, as a function of rotational speed, filling degree and mixing ratio for various types and mixtures of bed materials. These resulted in a relatively high limit value of $Fr = 10^{-2}$ for the transition to continuous rolling. Schnabel [11] developed a mathematical model for slumping that calculates the velocity and layer thickness of the slumping volume. The transition to rolling was, however, not recorded.

Prompted by the investigations [4,10,11], Henein et al. [6] conducted detailed studies of the motion behavior within the cascading range and devoted particular attention to the type of motion of slumping and the transition to

rolling. The calculation methods derived from this are taken up and further developed below.

3.3.1.1. Mathematical model for slumping. The cyclical process of elevation of the bed and slumping of a surface layer is subsequently denoted as slump cycle and is schematically depicted in Fig. 3. During the rotation of the cylinder the shear stress inside the slanting bed increases and reaches a critical value at an angle of inclination of the bed surface of α_U , the upper angle of repose. Near the bed surface a slip plane then forms, on which a thin particle layer begins to roll. The rolling of the particles and the removal of solids caused by this, however, occurs faster than the particle supply through the rotation of the cylinder. As a result, a temporal displacement of the slip plane occurs, decreasing its angle of inclination. The process of rolling ends when the angle of inclination of the slip plane has reached the value of α_L , the lower angle of repose. To understand the transition behavior of slumping–rolling, an analysis of the various processes within the slumping motion is worthwhile.

Slump cycle. A slump cycle may be subdivided into two phases: (a) elevation of the bed and (b) slumping of the bed material wedge (shear wedge). During the first step the lifting time t_1 is

$$t_1 = \frac{\pi \gamma_0}{180 \omega}. \quad (24)$$

In order to simplify the complicated flow of motion in the slumping phase, it is assumed that the entire rolling bed material, enclosed by the shear wedge angle γ_0 , suddenly slumps after reaching the upper angle of repose. The particles on the slope roll or slump in irregular motion and arrange themselves in the lower part of the bed, again filling in a shear wedge. Then the particles cover an average distance of s^* , which corresponds with the dis-

tance of the centers of gravity S_1 and S_2 of the shear wedges. On the average, the slip plane, the angle of inclination of which can assume values between α_U and α_L , is horizontally inclined by the angle η . If the cross-section of the shear wedge is regarded as a triangle ABC approximating the arc of a circle AB by a straight line (Fig. 3), then the distance of the center of gravity S_1 of point C corresponds with two thirds of half the chord of the bed. Thus, the length of the average particle path s^* can be calculated from

$$s^* = \frac{2}{3} s = \frac{4}{3} R \sin \varepsilon. \quad (25)$$

Owing to the smallness of the shear wedge angle γ_0 the centers of gravity S_1 and S_2 lie to a certain extent on its bisection, the angle of inclination η of the particle path. Hence, the latter can be calculated from

$$\eta = \frac{\alpha_L + \alpha_U}{2}. \quad (26)$$

In contrast to Eqs. (25) and (26), the quantities of s^* and η in [6] are calculated with great effort from the location of the centers of gravity S_1 and S_2 in a Cartesian coordinate system as a function of the angles α_U and α_L .

Slumping time t_2 . Due to lack of information on energy dissipation owing to particle collisions in the slumping step, it is assumed that the potential energy is converted solely to kinetic energy lowered by energy losses according to friction at the slip plane. Here the simple physical approach of solid body friction of the slumping volume at the slip plane is applied to model the frictional forces. Based on these assumptions, a trajectory model is developed, from which an equation for the slumping time t_2 is derived. Fig. 3 depicts the forces acting in the center of gravity S_1 of the slipping shear wedge. From the force balance parallel to the particle path $S_1 S_2$, the following ensues

$$F_G \sin \eta - F_I - F_F = 0. \quad (27)$$

The acceleration of gravity acts on the mass of a particle with

$$F_G = m_p g \quad (28)$$

and the kinetic energy is reflected in the force of inertia

$$F_I = m_p \frac{dv}{dt} = m_p \frac{d^2 x}{dt^2}. \quad (29)$$

Here v corresponds with the particle velocity along the path x with $0 \leq x \leq s^*$. According to the law of solid body friction, the following applies to the frictional force

$$F_F = \mu_i F_N. \quad (30)$$

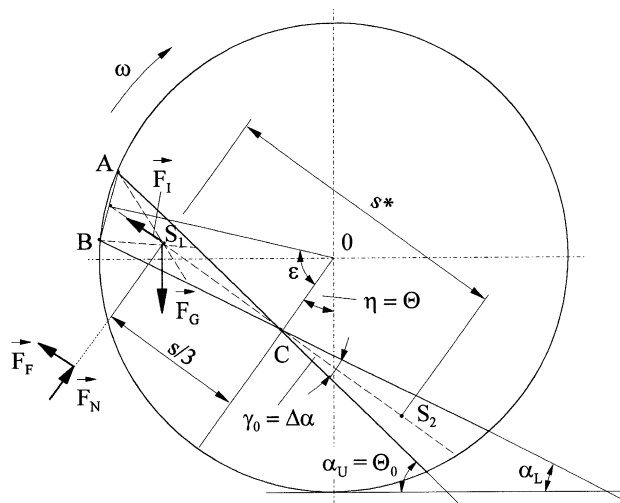


Fig. 3. Force balance around the center of gravity of the slipping shear wedge in the cross-section of a slumping bed.

In Ref. [6] the lower angle of repose α_L , which is also denoted as shear angle, is used as the angle of friction with

$\mu_i = \tan \alpha_L$. From the balance of forces perpendicular to the slip plane, the following ensues for the normal force

$$F_N = F_G \cos \eta. \quad (31)$$

After introducing Eqs. (28)–(31) in Eq. (27), a differential equation for the particle trajectory in the slumping phase results

$$\frac{d^2 x}{dt^2} = g(\sin \eta - \mu_i \cos \eta). \quad (32)$$

From the integration of Eq. (32) with the initial conditions $v(t=0) = 0$ and $x(t=0) = 0$, the path-time law of the slumping phase results

$$x = \frac{1}{2} g t^2 (\sin \eta - \mu_i \cos \eta). \quad (33)$$

After the slumping step is finished and the time t_2 has passed, the particles have covered the average distance s^* . With $x(t=t_2) = s^*$ and rearranging Eq. (33), the slumping time t_2 amounts to

$$t_2 = \sqrt{\frac{2s^*}{g(\sin \eta - \mu_i \cos \eta)}}. \quad (34)$$

3.3.1.2. Transition criterion. As the experiments of Henein et al. [6] revealed, slumping continues in a stable way as long as the shear wedge ABC can empty itself faster than it is filled anew (Fig. 3), i.e. as long as the slumping time is smaller than the lifting time, that is $t_2 < t_1$. As rotational speed increases, however, the lifting time t_1 decreases more than the slumping time t_2 and the measured curve of $t_1(n)$ approaches the progression of $t_2(n)$. Afterwards, both phases of the slump cycle above a certain rotational speed range are in equilibrium until a critical rotational speed is reached at which slumping turns into rolling. The equilibrium between filling and emptying of the shear wedge can therefore be regarded as a critical state for the transition to continuous rolling. If the condition

$$t_1 < t_2 \quad (35)$$

is met, continuous rolling occurs. This condition is used to obtain a criterion for the transition from slumping to rolling. Using (Eqs. (24), (25) and (34), the critical Froude number for this transition results from the condition (35) to

$$Fr_c = \frac{3}{8} \left(\frac{\pi \gamma_0}{180} \right)^2 \frac{\sin \eta - \mu_i \cos \eta}{\sin \varepsilon}. \quad (36)$$

Besides the filling degree, the criterion (36) in combination with Eq. (26) is dependent on specific bed material properties, which characterize the behavior of motion in the case of periodic slumping, such as the lower and upper angle of repose α_L and α_U , respectively, and the shear wedge angle γ_0 . The measurement of these bed material properties is, however, only possible with great experimental effort, Eq. (36) being usable only to a limited extent.

In order to transform this criterion into an applicable form, simplifications regarding the bed material properties are made. As measurements from Henein et al. [6] demonstrate, the angle η calculated from Eq. (26) is nearly identical with the dynamic angle of repose Θ , that is $\eta \approx \Theta$. In addition, the upper angle of repose α_U corresponds to a large extent with the static angle of repose Θ_0 ($\alpha_U \approx \Theta_0$) and the measured shear wedge angle is nearly $\gamma_0 \approx \Delta \alpha = \alpha_U - \alpha_L$. The shear wedge angle thus approximately results from

$$\Delta \alpha = 2(\Theta_0 - \Theta). \quad (37)$$

Using the aforementioned simplifications the following results from Eq. (36)

$$Fr_c = \frac{3}{2} \left[\frac{\pi(\Theta_0 - \Theta)}{180} \right]^2 \frac{\sin \Theta - \tan(2\Theta - \Theta_0) \cos \Theta}{\sin \varepsilon}. \quad (38)$$

Hence, as is clear from Eq. (38), the slumping–rolling transition may be specified as a function of the flow properties used in particle technology—static and dynamic angles of repose. Fig. 4 depicts the progression of the critical Froude number in terms of the filling degree for three different solids: gravel, limestone, and sand. The bed material properties of these (Table 2) were obtained from Ref. [6] with the exception of sand; its values were taken from this author's own data using sand with a similar particle diameter ($d = 0.5$ mm) because the dynamic angle of repose of sand given in Ref. [6] was greater than the static angle of repose.

It is obvious that the fine grained, free flowing sand already begins continuous rolling at a Froude number of approximately $Fr = 2 \cdot 10^{-5}$. By comparison, the transition for the coarser grained gravel is higher by more than a decimal power. As the graph shows, the predicted curves for gravel and sand compare well with the data. However, the calculated progression for limestone deviates from the measured values by approximately half a decimal power. The reason for these differences possibly lies in measuring errors when determining the flow properties. For example, when the static angle of repose Θ_0 deviates by 1° , the Froude number varies by the 3- to 6-fold. A further cause is the assumptions of the simplified mathematical model used.

In addition, Fig. 4 contains the predictions from the slipping criterion derived by Henein et al. [6]. A direct comparison between this criterion and Eq. (38) is impossible because of the above-mentioned differences in the calculation of the quantities s^* and η . Moreover, values of these quantities are not named. Despite this circumstance, calculations of the slipping criterion were performed using relationships (25) and (26). While the criterion of Henein et al. calculated in this way compares well with the measurements of limestone, it deviates significantly from

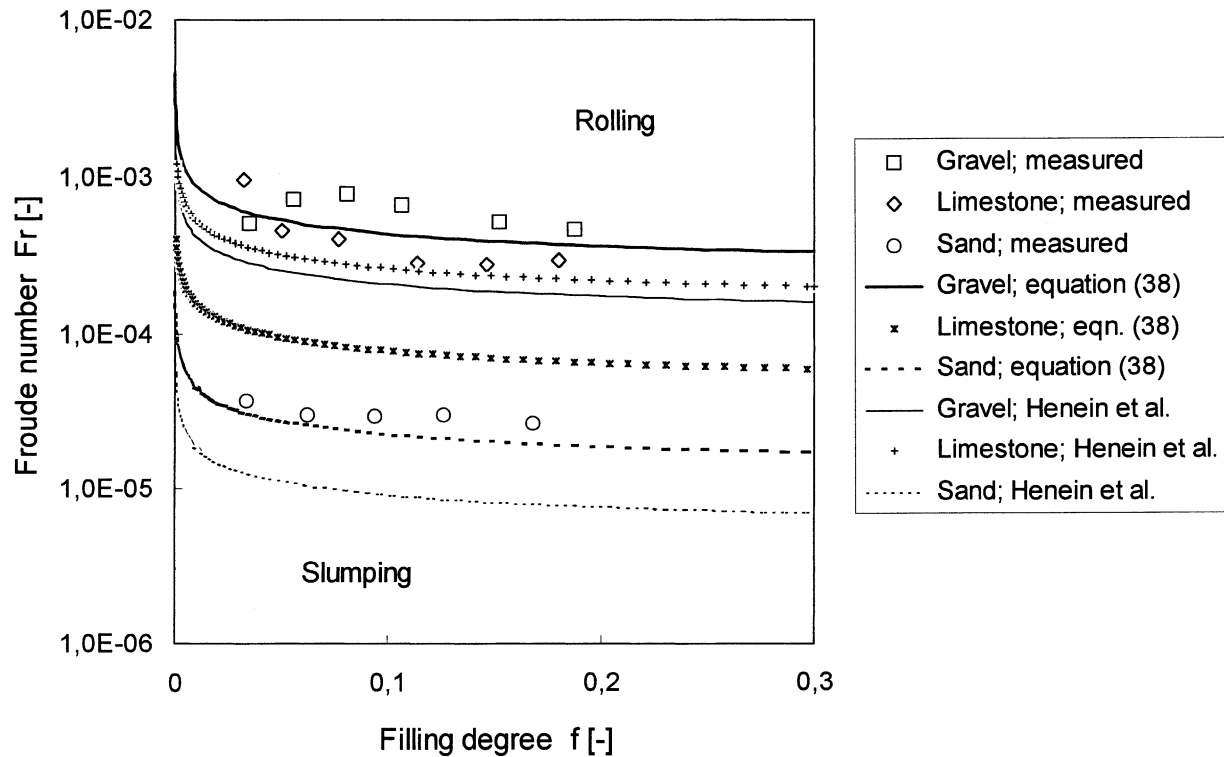


Fig. 4. Transition slumping–rolling: critical Froude number in terms of filling degree for different bed materials in comparison with data and calculations from Henein et al. [6].

those of gravel and sand. As the comparisons make clear, the simple physical model developed above describes the transition from slumping to rolling with sufficient accuracy.

3.3.2. Transition of rolling–cascading

Up to now the transition between rolling and cascading has been studied only a little [6,8]. The nearly flat bed surface is a characteristic feature of rolling. The significant curvature of the bed is named in the literature as a feature of the transition to cascading. However, mathematically specifying the flatness of an agitated bed surface is difficult. Apart from that, subjective influences strongly characterize its experimental determination. According to Henein et al. [6] cascading occurs when the filling angle exceeds a critical value of $\varepsilon = 90^\circ - \Theta$ and places the apex of the bed in the II quadrant (Fig. 5). As investigations of Blumberg [8] and the author's own experiments demonstrated, cascading can also occur at lower filling degrees. According to Blumberg's experiments [8], besides

Table 2
Physical characteristics of gravel, limestone [6], and sand

Material	Particle diameter d [mm]	Dynamic angle of repose Θ [°]	Static angle of repose Θ_0 [°]
Gravel	3.0	37.5	40.7
Limestone	1.5	36.0	37.8
Sand	0.5	32.5	33.7

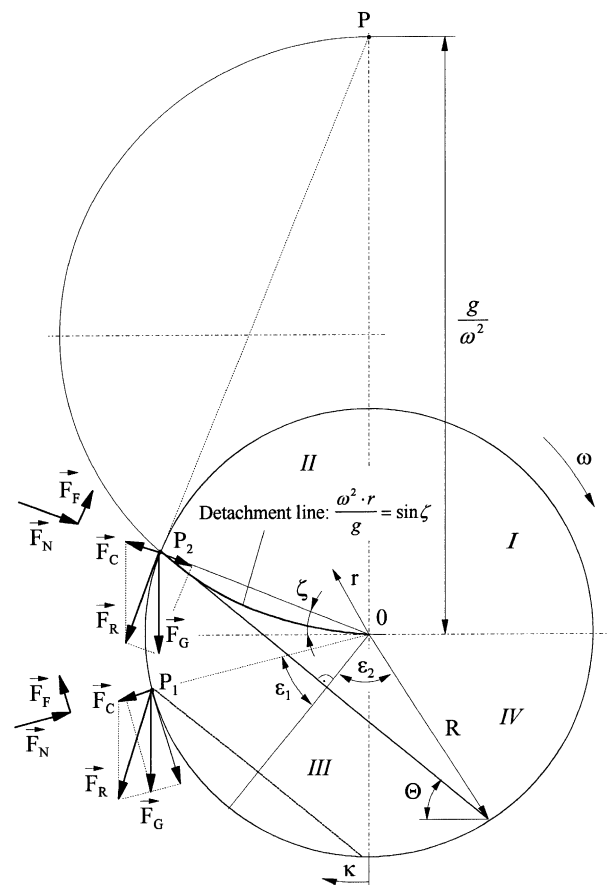


Fig. 5. Force balances around particles in different quadrants of the cylinder cross-section.

the rotational speed, this transition is also dependent on particle size. The following criterion for the rolling–cascading transition was ascertained here through observations from experiments on rotating drums with transparent front plates

$$Fr\left(\frac{D}{d}\right) \approx 2, \quad (39)$$

the validity of which, however, is not proven for significantly larger dimensions [8]. In order to validate Eq. (39) and to attain data on the transition from rolling to cascading, the author carried out experiments, which are described in Section 4.

3.4. Transition of cascading motion–cataracting motion

In the literature, the throw off of particles into the free gas space is generally named as the characteristic for the transition from cascading motion to cataracting motion. Since the introduction of ball mills in materials processing technology more than 100 years ago [22], the motion behavior of the bed in this rotational speed range has been already researched in numerous studies, among others in Refs. [6,12–14,22–24]. A transition criterion produced by Henein et al. [6] is based on the condition that particles thrown off from the apex of the bed fall to, or beyond, the mid-point of the slope. This assumption requires the calculation of the particle trajectories. Diedrich [12] conducted experiments to determine the trajectories of the balls in a model ball mill and derived equations for their calculation. Mu and Perlmutter [17] also specified equations for particle trajectories. Teubner [13] analyzed the motion behavior of particles in various quadrants of the rotary kiln on the basis of force balances. He then established that though centrifuging theoretically begins with a Froude Number of $Fr = 1$, for the adhesion of a particle on the rotary kiln wall (depending on the position on the circumference), a multiple of this Froude number may, however, be required ($Fr > 1$). Taking into account the gravity and the centrifugal force, Davis [23] applied the radial equilibrium of forces on a particle at the cylinder wall as condition for the throw off into the free gas space. Considering particles at different orbits in the bed, the radial equilibrium of forces results in the line of detachment, which obeys a circle through the axis of rotation. The diameter of this circle is g/ω^2 , the perpendicular distance of the pole of forces of the gravitational force and the centrifugal force from the rotational axis. Including the frictional force, Zengler [14] considered the radial balance of forces on particles projected from the bed surface. Then the line of detachment has the form of a logarithmic spiral, the central point of which is located in the pole of forces. According to Rutgers [3], the transition between cascading motion and cataracting motion lies in the range of the Froude number of $Fr = 0.3$ – 0.36 .

3.4.1. Transition condition

The detachment of particles due to the radial equilibrium of forces [9,13,23], which according to Davis [23] is considered to be the minimum condition for the throw off of particles into the gas space, is used as criterion for the transition to cataracting motion in the following. For this purpose, force balances on particles at different positions in the cross-section of a rotating cylinder are worked out; see points P_1 and P_2 in Fig. 5. The gravitational force F_G , the centrifugal force F_C and the frictional force F_F act on every particle in the plug flow region, which moves in an orbit around the axis of rotation. The resulting force F_R is produced by the interaction between the gravitational and centrifugal forces. The lines of influence of all resulting forces in the plug flow region intersect in the pole of forces P . If the centrifugal force and the component of the gravitational force which is directed inward radially cancel each other out, as in point P_2 , then radial equilibrium of forces exists, from which the equation of the line of detachment results

$$\frac{\omega^2 r}{g} = \sin \zeta. \quad (40)$$

According to Thales' Principle, the line of detachment describes a circle with the diameter g/ω^2 , which corresponds to the perpendicular distance of the pole of forces from the axis of rotation OP . This line runs through the II quadrant of the cross-section of the rotating cylinder (in the mathematically positive direction of rotation). Eq. (40) specifies at which angle ζ , as a function of the orbital radius r , a particle begins to detach. That means, if the particle paths do not reach this line, cascading motion occurs. If the line is exceeded, cataracting motion appears. It is obvious from this that particles can throw off only in the II quadrant. For particle throw off, however, the condition

$$\frac{\omega^2 r}{g} > \sin \zeta \quad (41)$$

must be met. The progression of the line of detachment also makes clear that particles in the proximity of the wall are thrown off first. Hence, only particles in the outermost orbit $r = R$ are considered below. Force balances are worked out on these particles, from which criteria for the transition of cascading motion to cataracting motion are derived. In addition, the different directions of action of the forces in the upper and lower parts of the rotating cylinder cross-section are taken into consideration. Depending on the angle

$$\zeta = \Theta + \varepsilon - 90^\circ = \kappa - 90^\circ, \quad (42)$$

the solid is either completely in the lower section (III and IV quadrant) or it extends as far as the I and II quadrants when the filling degrees are greater. Consequently, when

ascertaining the transition behavior two cases must be distinguished: low filling degrees ($\Theta + \varepsilon < 90^\circ$) and higher filling degrees ($\Theta + \varepsilon > 90^\circ$). As will be shown, only the region of higher filling degrees is relevant in practice.

3.4.2. Low filling degrees: $\Theta + \varepsilon < 90^\circ$ ($\zeta < 0$)

Particles which move in an orbit below the axis of rotation are prevented from throw off into the gas space. Here, just as much as the centrifugal force, the vectors of the gravitational force components are radially oriented towards the outside so that the particles are pushed against the rotating wall (see point P_1 in Fig. 5). Only when the rotational speed is higher and the wall roughness is sufficiently greater can a particle transport into the II quadrant through the adhesive friction of particles on the wall occur, in which the throw off is possible. At an angle of $\kappa = \Theta + \varepsilon = 90^\circ$, the contact force necessary for this must be supplied by the centrifugal force alone. The corresponding critical Froude number is ascertained below. In tangential direction, the force balance around the particle in point P_1 amounts to

$$m_p g \sin \kappa = \mu_w m_p (g \cos \kappa + \omega^2 R). \quad (43)$$

Here μ_w is the coefficient of friction between particle and cylinder wall. If the particle is elevated by the angle $\kappa = 90^\circ$, it follows that $\mu_w Fr = 1$. However, the condition

$$\mu_w Fr > 1 \quad (44)$$

must be met in order to transport the particle into the II quadrant and ensure throw off. This condition is applied below as criterion for reaching cataracting motion when the filling degree is low. Strictly taken, it applies only to a single particle, since, besides the wall friction, the inner bed material friction is important in the motion of a bed. The critical Froude number is thus calculated from

$$Fr_c = \frac{1}{\mu_w} \quad (45)$$

and is inversely proportional to the coefficients of friction of particle/wall or bed material/wall, respectively. The progression of Eq. (45) is visible from Fig. 6. The graph makes clear that very high rotational speeds ($Fr > 1$) would be necessary for the transition of cascading motion to cataracting motion when the filling degree is low ($\zeta < 0$) and in practice usual wall friction coefficients for technical bed materials of $\mu_w < 1$ occur. For example, the wall friction angle for sand in a rotary kiln with sintering zone roughness amounts to approximately $\varphi_w = 34^\circ$ [25], from which a coefficient of wall friction of $\mu_w = 0.675$ results. Therefore, the critical Froude number amounts to $Fr_c = 1.48$. Above this Froude number, it would be possible for this bed material to reach the II quadrant and thus cataracting motion. With a dynamic angle of repose of $\Theta = 32.5^\circ$, the limit filling angle for sand amounts to $\varepsilon(\zeta = 0) = 57.5^\circ$. That means, under conditions of cascading motion, the bed of sand is in the lower section of the rotary kiln up to a

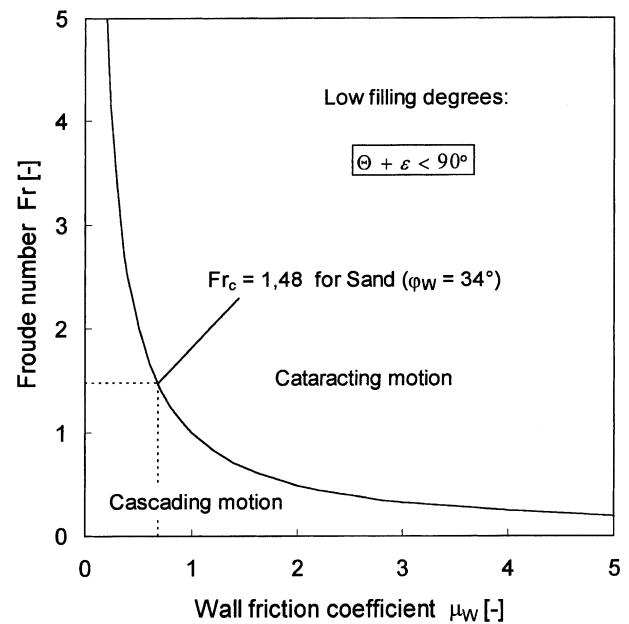


Fig. 6. Transition cascading motion–cataracting motion at low filling degrees: critical Froude number in dependence on wall friction coefficient.

filling degree of 17.5%. Actually the transition even occurs at somewhat smaller filling degrees, since the bed is kidney-shaped in cascading, so that a part of the bed rises up into the II quadrant.

It is, however, known [3,27] from experiments and practical experience with ball mills that cataracting motion is already achieved with Froude numbers in the range of 0.25–0.72 and these rotating cylinders are operated with higher filling degrees.

3.4.3. Higher filling degrees: $\Theta + \varepsilon > 90^\circ$ ($\zeta > 0$)

Particles which move in orbits in the upper part of the cylinder cross-section—i.e. in the I and II quadrants ($\zeta > 0$)—detach from the bed and can be thrown off if radial equilibrium of forces exists. From the balance of forces acting on a particle in the point P_2 (Fig. 5), the normal force amounts to

$$F_N = F_C - F_G \sin \zeta. \quad (46)$$

For slow rotation the normal force F_N becomes negative. In this case a solid particle elevated by the cylinder wall would roll (cascading motion) on the bed surface after reaching point P_2 . The radial equilibrium of forces is reached, if the normal force approaches zero. For the throw off of this particle, in accordance with relationship (41), the condition

$$\frac{\omega^2 R}{g} > \sin \zeta \quad (47)$$

must be met. Converting inequality (47) into an equation, the critical Froude number for the transition of cascading

motion to cataracting motion with higher filling degrees amounts to

$$Fr_c = \sin(\Theta + \varepsilon - 90^\circ). \quad (48)$$

Hence, the critical Froude number depends on the filling degree and the dynamic angle of repose of the bed material. Fig. 7 plots the progression of Eq. (48) against the filling degree with dynamic angles of repose in the range of $25^\circ \leq \Theta \leq 40^\circ$. As the graph makes clear, a detachment of the particles is possible when filling degrees are approximately above 15%. The lower the dynamic angle of repose, the higher the filling degree necessary to reach cataracting motion. To a large extent the results of the model calculations correspond with practical experiences from ball mills [3,6], according to which cataracting motion is obtained with Froude numbers in the range of $Fr = 0.25\text{--}0.72$. Ball mills are operated in the transition range between cascading motion and cataracting motion. Actual operating parameters, however, are filling degrees between 0.35 and 0.50 and rotational speeds of up to 65–75% of the critical speed, that is $Fr = 0.42\text{--}0.56$ [27]. This range of operating parameters of ball mills is depicted in Fig. 7 (hatched area), emphasizing a good correspondence between theory and practice. Here the dynamic angle of repose of the mixture of ball charge and bed material to be grinded must be applied. The predictions also agree with a characteristic for the transition of cascading motion to cataracting motion presented by Rose and Sullivan [28], a graph of the critical filling degree in terms of the rotational speed $f = f(n/n_c)$.

3.5. Transition of cataracting–centrifuging

Studies on the transition behavior within the range of cataracting motion are presented in Refs. [9,13,24] among

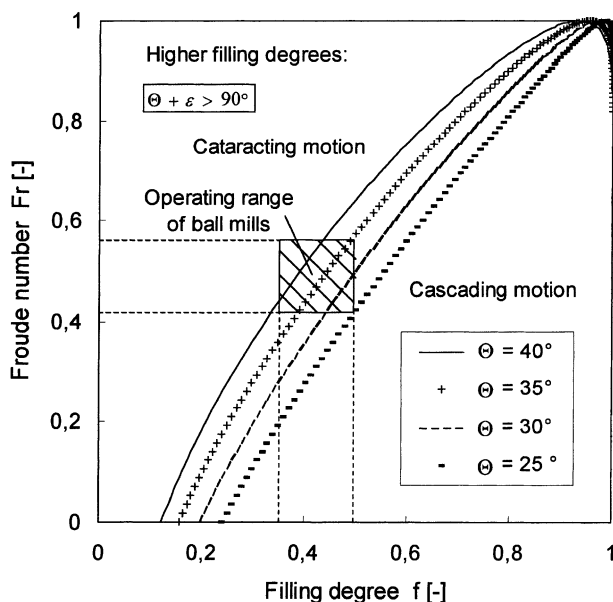


Fig. 7. Transition cascading motion–cataracting motion at higher filling degrees: critical Froude number in terms of filling degree and dynamic angle of repose, compared with operating parameters of ball mills [27].

others. Kelbert and Royere [1] name altogether five sub-types of cataracting motion, from which Korotich [9] investigates the types of motion “waterfall type” and “circulatory type” and applies an empirical criterion to the transition between these. Teubner [13] derives an equation for the centrifuging of a single particle from the force balance on a particle, which moves in the orbit $r = R$ around the axis of rotation. Watanabe [24] develops a mathematical model for the transition behavior of the ball charge between cataracting and centrifuging on the basis of the Discrete Element Method (DEM), the results of which agree well with measurements on a model ball mill.

According to the classic definition of centrifuging, this state of motion is attained by particles, which move in orbits around the axis of rotation, if the forces, centrifugal force and gravitational force, acting on them cancel each other out. This radial equilibrium of forces exists in the upper dead center of the rotating cylinder when $\kappa = 180^\circ$, in which the lines of influence of both forces run parallel (Fig. 5). Thus, from Eq. (40), the known critical Froude number for the transition of cataracting–centrifuging amounts to

$$Fr = 1. \quad (49)$$

Far higher Froude numbers $Fr > 1$ are, however, required for the adhesion of a particle on the drum wall with smaller angles of deflection $\kappa < 180^\circ$ (or higher angles of deflection $\kappa > 180^\circ$). This connection becomes obvious if the tangential balance of forces is considered, in which, in contrast to the radial force balance, the frictional force at the wall F_F is additionally taken into account (see Fig. 5). The following results from the tangential equilibrium of forces

$$Fr = \frac{\sin \kappa}{\mu_w} - \cos \kappa = \frac{\sin(\kappa - \varphi_w)}{\sin \varphi_w}, \quad (50)$$

$\mu_w = \tan \varphi_w$ characterizing the effective coefficient of particle/wall friction. Using this relationship, a Froude number is calculated, which, with a given coefficient of friction, is required in order to hold a particle on the rotating wall up to an angle of deflection of κ . While Eq. (49) is only valid at the angle $\kappa = 180^\circ$, Eq. (50) calculates the critical Froude number for a single particle in any position κ on the perimeter. Fig. 8 illustrates its curve progression for various coefficients of wall friction. As expected, under the angle of $\kappa = 90^\circ$, Eq. (50) turns into criterion (45) for the transition of cascading motion–cataracting motion at low filling degrees. Furthermore, with extremely high wall friction coefficients $\mu_w \rightarrow \infty$ and $\kappa = \Theta + \varepsilon$, Eq. (50) delivers the criterion (48) for higher filling degrees. As the graph shows, when $\kappa = 90^\circ + \varphi_w$, the critical Froude number for a single particle reaches its maximum with

$$Fr_{c,p} = \frac{1}{\sin \varphi_w}. \quad (51)$$

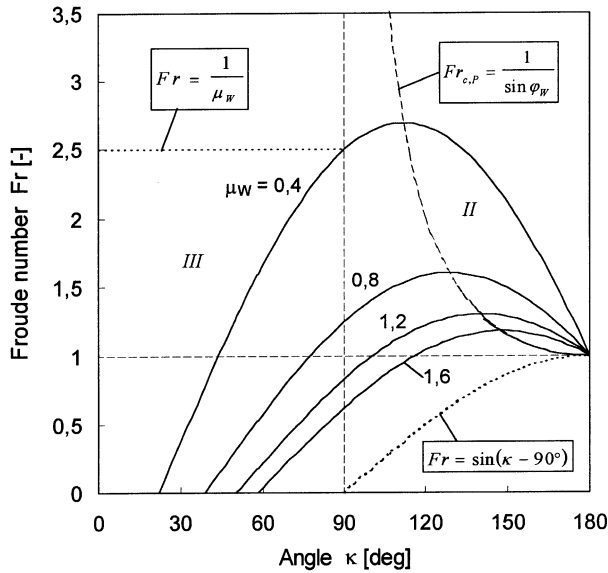


Fig. 8. Critical Froude number of centrifuging of a single particle in terms of its angular position and wall friction coefficient.

According to Teubner [13], the relationship (51) specifies the “actual” critical Froude number for the cataracting–centrifuging transition, because above this every particle on the perimeter is centrifuged. For example, with a coefficient of wall friction of $\mu_w = 0.4$, the Froude number reaches its maximum value of $Fr = 2.7$ under an angle of $\kappa = 111.8^\circ$. These ideas find their confirmation in experiments on ball mills [12,13], which prove that in the experiment centrifuging only begins at higher rotational speeds ($Fr > 1$). Even if Eq. (51) represents the criterion for centrifuging of only a single particle, conclusions about the behavior of a bed of particles can be drawn from its progression.

In the case of cataracting motion in a ball mill, the center of gravity of the ball charge is deflected horizontally by at least the dynamic angle of repose Θ . At sufficiently high filling degrees, the apex of the bed with $\kappa = \Theta + \varepsilon$ may exceed the critical angle $\kappa = 90^\circ + \varphi_w$ for centrifuging of a single particle. Under this condition the motion of the bed can turn into centrifuging if the Froude number, in accordance with Eq. (51), exceeds the critical value when $\varphi_w = \Theta + \varepsilon - 90^\circ$

$$Fr_c = \frac{1}{\sin(\varphi_i + \varepsilon - 90^\circ)}. \quad (52)$$

Instead of the dynamic angle of repose Θ , which characterizes the bed behavior in cascading motion, the angle of internal friction of the ball charge φ_i is applied in Eq. (52), being valid if the ball diameter is negligibly small in comparison with the cylinder diameter. Besides the angle of internal friction, the critical Froude number is dependent on the filling degree. Naturally, Eq. (52) is only valid for higher filling degrees $\varphi_i + \varepsilon > 90^\circ$. More recent studies by Watanabe [24] on a model ball mill ($L = 0.124$ m; $D =$

0.76 m) confirm the strong influence of the filling degree. As the filling degree increases, the Froude number sharply decreases and asymptotically approaches the value $Fr = 1$, which is achieved with a filling degree of approximately 70%. Fig. 9 compares these studies with Eq. (52), which shows a tendential correspondence with the measurements. Despite the deviations from the measured values for a friction coefficient of $\mu_i = \mu_w = 0.15$ specified in Ref. [24], this relationship is used below as approximation method to calculate the transition of cataracting–centrifuging.

Watanabe considers the critical rotational speed to be reached as soon as the outer particle layer forms a ring. Simulation calculations following the Discrete Element Method (DEM) using the linear–spring–dashpot model for particle–particle and particle–wall collisions as well as the Coulomb criterion for dynamic friction produced a good correspondence with the experimental results [24]. No centrifuging was observed when filling degrees were less than 30%. If differences in the void volume between resting ball charge and centrifuged ball ring are disregarded, the minimum filling degree of a ball mill necessary for the formation of a closed ball ring of the thickness $R - R_i = d$ is produced from the ratio of ball to drum diameter using

$$f_{\min} = 1 - \left[1 - 2 \left(\frac{d}{D} \right) \right]^2. \quad (53)$$

Ring formation is impossible at filling degrees less than $f_{\min}$, as a result of which centrifuging becomes unstable in this range. The minimum filling degree for the diameter ratio $d/D = 0.066$ selected in Ref. [24] is plotted in Fig. 9. It is obvious that the progression of the critical Froude

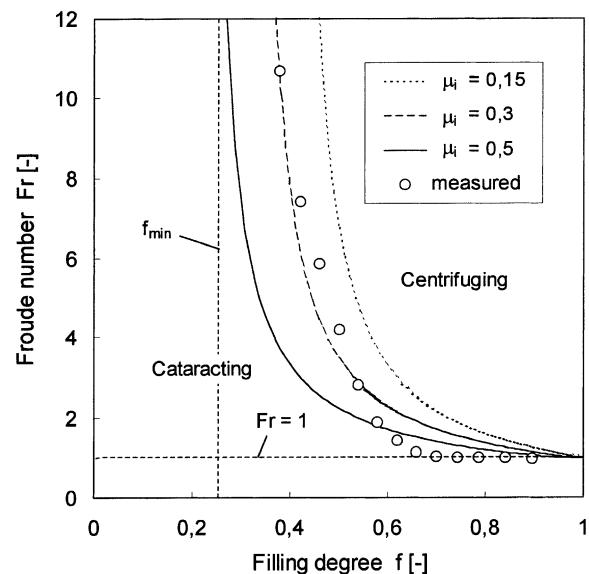


Fig. 9. Transition cataracting–centrifuging: calculated boundaries in terms of filling degree and coefficient of internal friction in comparison with data from Watanabe [24].

number according to the data and Eq. (52) approaches the minimum filling degree asymptotically as the filling degree decreases.

4. Experiments

4.1. Transition of slipping motion–cascading motion

During previous residence time studies on a rotating cylinder of glass ($L = 4.6$ m; $D = 0.3$ m), the transition from slipping motion to cascading motion has been observed as a side effect [25]. Using quartz sand ($d = 0.27$ mm; $\Theta = 32.4^\circ$) as bed material, the residence time and the filling degree were ascertained in dependence on the solids mass flow rate in these experiments. Through the transparent cylinder wall, the axial progression of the filling degree has been measured, from which the mean residence time was calculated. When observing the solids transport through the glass cylinder, a discontinuity in the transverse bed motion was detected. This discontinuity expressed itself in the residence time functions which, plotted against the solids mass flow rate, exhibit a typical break in small mass flows. As the mass flow rate increased, the residence time then decreased, to slightly increase again after the breaking point. Small mass flow rates result in a low filling degree in the rotary kiln, which, in case of a smooth wall, may cause slipping motion. For low filling degrees, it could be observed that the entire bed in the cylinder moves in slipping motion and, as filling degree and mass flow rate increase, turns into cascading motion after the breaking point. A complete set of readings is depicted in Fig. 10, which is a plot of the dimensionless

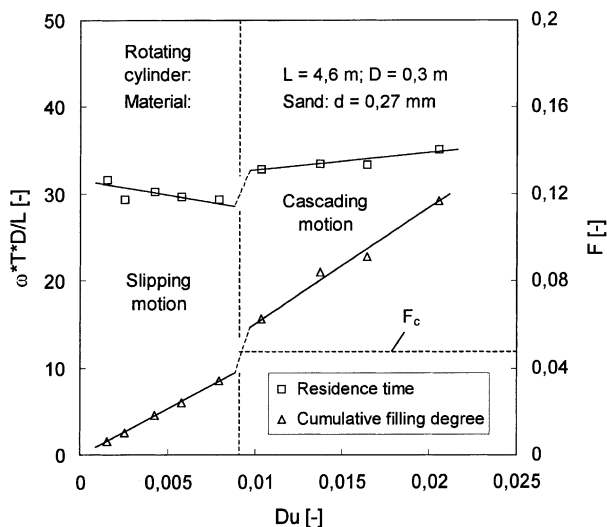


Fig. 10. Dimensionless functions of the residence time and the cumulative filling degree in terms of solids mass flow rate measured in a rotating cylinder of glass using quartz sand [25].

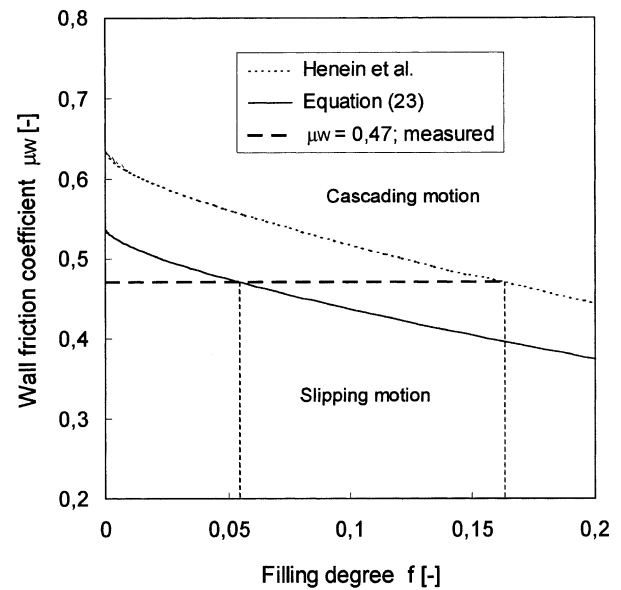


Fig. 11. Transition slipping motion–cascading motion: critical wall friction coefficients calculated in terms of the filling degree in comparison with data of quartz sand/glass measured in a Jenike shear tester.

residence time $\omega T(D/L)$ and the cumulative filling degree F against the dimensionless solids mass flow rate

$$Du = \frac{\dot{m}_s}{\rho_b \omega R^3}. \quad (54)$$

As the graph shows, both the residence time and the filling degree curves exhibit typical breaks at a dimensionless mass flow rate of $Du \approx 0.009$. The critical filling degree at the breaking point amounts to $0.04 < F_c < 0.06$.

In order to use these experimental results for comparison with model predictions, wall friction angles of quartz sand/glass were measured in a Jenike shear tester at low static pressures less than 2 kPa. These measurements gained data in the range of $\varphi_w = 22.7\text{--}26.5^\circ$ resulting in a mean wall friction coefficient of $\mu_w = 0.47$. In Fig. 11, this value is compared with the critical wall friction coefficients calculated from Eq. (23) and from the slipping criterion of Henein et al. [6]. As the graph shows, the criterion of Henein et al. produces about 19% higher values due to the tangent of the dynamic angle of repose used instead of sine, which is applied in Eq. (23). In comparison with the measured wall friction coefficient, this criterion calculates a critical filling degree of $f_c = 0.165$ whereas the curve predicted from Eq. (23) intersects the μ_w -line at a value of $f_c = 0.054$, which corresponds well with the critical filling degree estimated from the residence time series. That is, if the burden of the bed material exceeds the critical region of $0.04 < f_c < 0.06$, the wall friction coefficient measured is sufficient to ensure cascading motion. This comparison also makes clear that the assumptions adopted from Henein et al. are not entirely correct. Hence, the mathematical model for the transition from slipping motion to cascading motion developed in this study adequately describes this phenomenon. How-

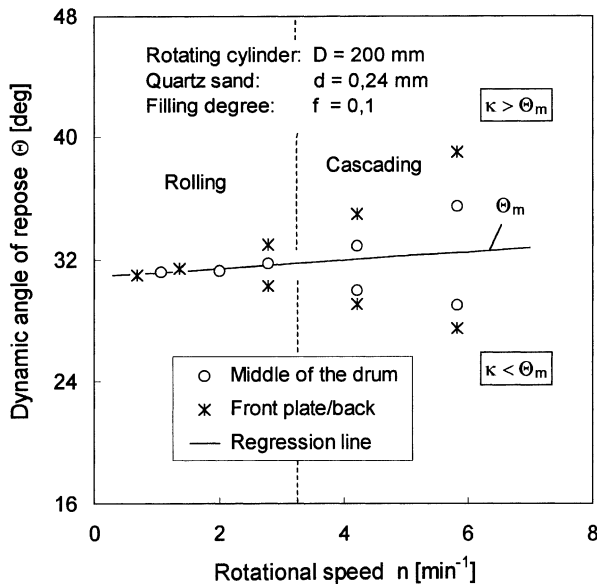


Fig. 12. Dynamic angle of repose of quartz sand measured in different segments of the slope κ in dependence on rotational speed.

ever, further experiments are necessary to prove this criterion.

4.2. Transition of rolling–cascading

Experimental research has been conducted in order to ascertain the transition from rolling to cascading by measuring the dynamic angle of repose Θ of the bed in rotating cylinders with glass front plates. Various quartz sand fractions of narrow particle size distributions with mean particle diameters of $d = d_{50} = 0.24; 0.5; 0.75; 1.0$ mm were used. Besides the particle diameter, the variables of rotational speed ($0.5\text{--}10 \text{ min}^{-1}$), filling degree ($0.05\text{--}0.25$), and cylinder diameter (200, 300, 500 mm) were varied. In each test run, photographs of the agitated bed were taken through the glass plate. In order to measure the filling degree and the dynamic angle of repose, a metric scale and a plumbline were installed in front of the drum.

In the rolling range, the angle of inclination of the bed surface corresponds to the dynamic angle of repose Θ , which was found to be slightly dependent on rotational speed and filling degree. A linear relationship between the dynamic angle of repose and the rotational speed was established. As the rotational speed increases, cascading appears, resulting in a kidney-shaped bed of different inclination angles of the bed surface in different segments of the slope. Taking into consideration the pulling effect of the side walls, these inclination angles were measured both at the front plate/back and in the axial middle of the drum, i.e. in the undisturbed flow region of the bed, and compared with each other. For this purpose, the photographs were taken from the top view and the slanting top view as well. To obtain an undisturbed particle flow in the middle of the drum, cylinders of a length of $L/D > 0.3$ were used.

Fig. 12 depicts measurements of quartz sand ($d = 0.24$ mm) rotated in a cylinder of 0.2 m diameter that was filled 10%. It is obvious that, as the rotational speed increases, the inclination angles of the slope in the different segments of the bed surface $\kappa < \Theta_m$ and $\kappa > \Theta_m$ clearly increase or decrease from the point of transition, respectively. Owing to the pulling effect, the transition behavior at the side walls deviates from that of the undisturbed flow region. While the critical speed of rotation measured at the front plate/back is approximately $n_c = 2 \text{ min}^{-1}$, the value for the undisturbed particle flow amounts to $n_c = 3.3 \text{ min}^{-1}$. Within the cascading range, the measured angles can be approximated by a straight line (Θ_m) which follows the gradient of the dynamic angle of repose Θ in the rolling range.

Fig. 13 compares a complete set of readings with the criterion of Blumberg [8] according to relationship (39). This diagram depicts the critical Froude number for the rolling–cascading transition, divided by the ratio of particle to cylinder diameter as a function of filling degree. As the graph shows, the critical Froude number measured in the undisturbed flow is higher than that at the front plate and the back. The deviations from the experimental values allow the presumption of a dependence of the critical Froude number on operating variables, which are not accounted for in Eq. (39). As it appears, besides the rotational speed and particle diameter, the transition in the area of parameters studied is also dependent on filling degree. In all experiments the transition rolling–cascading estimated in this way was in the range of $Fr = 10^{-4}\text{--}10^{-2}$. The cylinder diameter had no influence. Despite the deviations from the measurements and for lack of a suitable

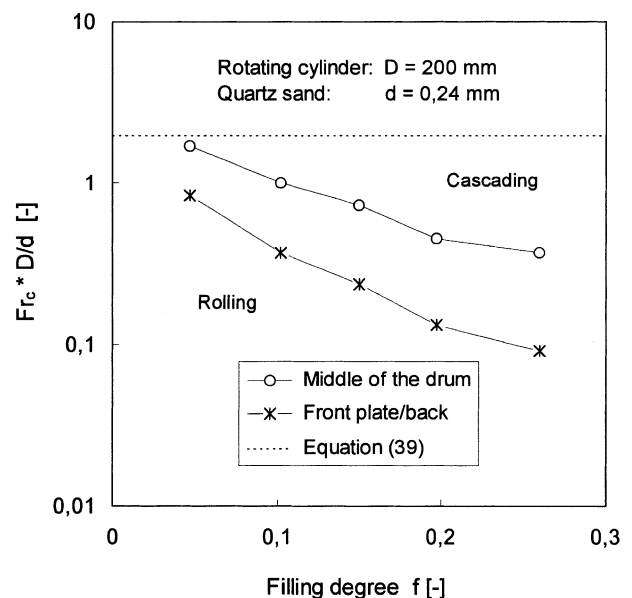


Fig. 13. Transition rolling–cascading: data of the critical Froude number of quartz sand as $Fr_c D/d$ vs. filling degree in comparison with the criterion of Blumberg [8].

mathematical model, Eq. (39) is used below to carry out the calculations for the Bed Behavior Diagram.

5. Bed behavior diagram

The transverse motion behavior of granular solids in rotating cylinders can be conveniently represented on a Bed Behavior Diagram, a similar form of which was already proposed by Henein et al. [6]. The Bed Behavior Diagram used in this paper plots the Froude number against the filling degree, in which the ranges of the individual forms of motion and their limits are illustrated. The transition between slipping motion and cascading motion constitutes an exception, in which, in comparison with the gravitational force, the centrifugal force is usually

negligibly small and the wall frictional force is determining; here the coefficient of bed material/wall friction is plotted as a function of the filling degree.

Figs. 14–16 present complete Bed Behavior Diagrams for three different bed materials (gravel, limestone, sand). From the physical characteristics of the bed materials indicated in Table 2, the boundaries between the types of motion were calculated using the following criteria:

slipping motion–	Eq. (23)
cascading motion	
slumping–rolling	Eq. (38)
rolling–cascading	Eq. (39)
cascading motion–	Eqs. (45) and (48)
cataracting motion	
cataracting–centrifuging	Eq. (52).

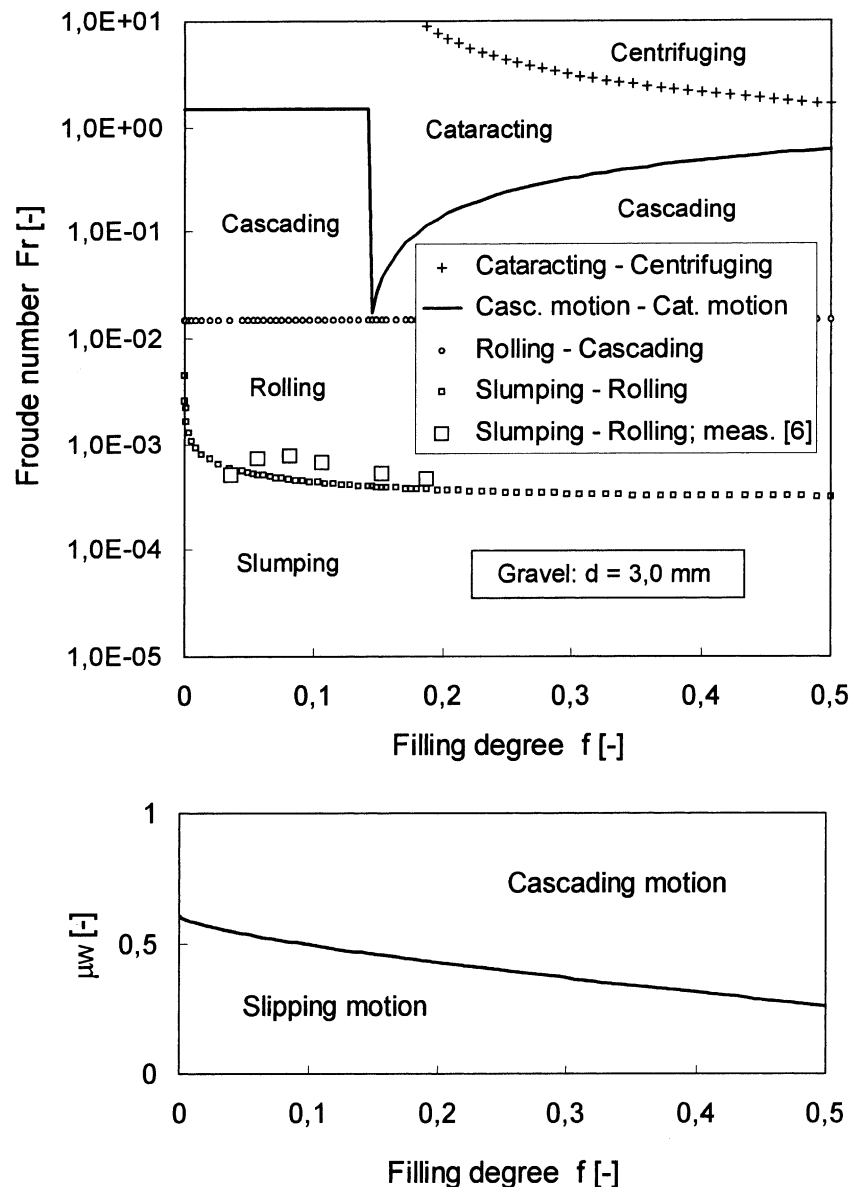


Fig. 14. Bed Behavior Diagram of gravel.

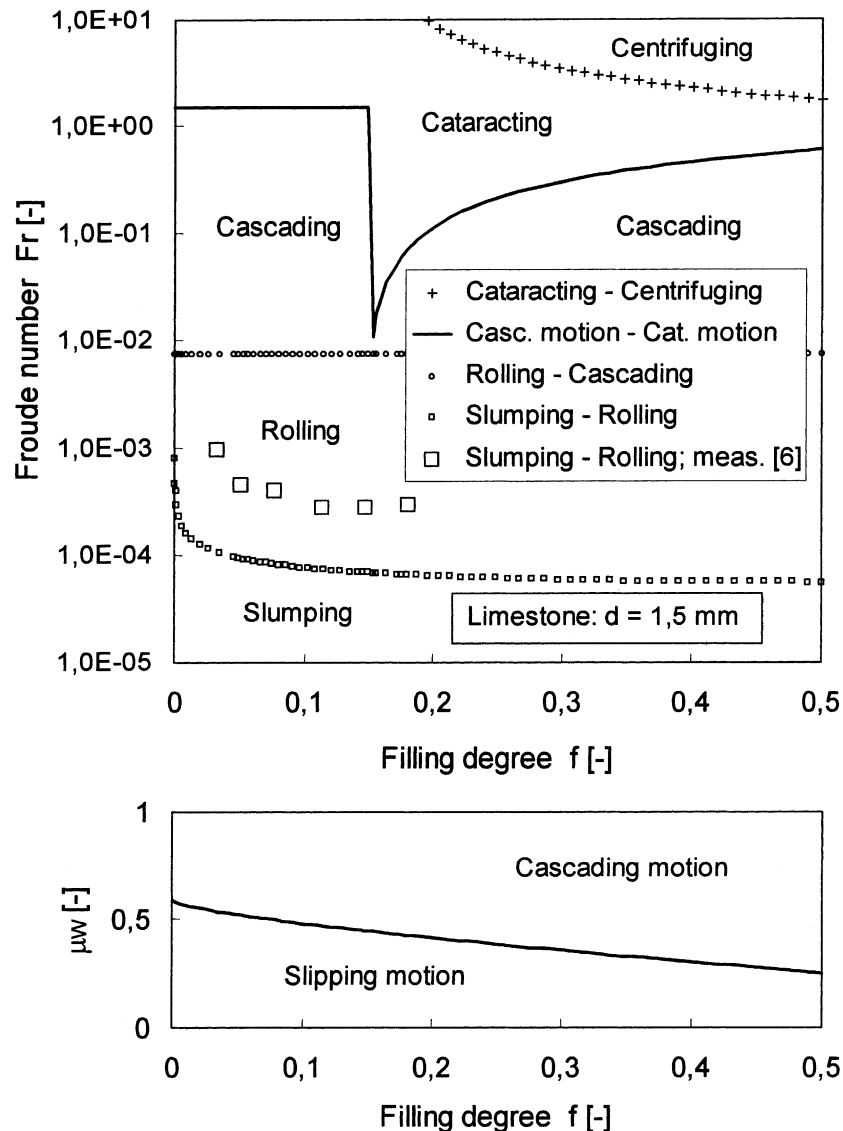


Fig. 15. Bed Behavior Diagram of limestone.

The transition between slipping motion and cascading motion calculated from Eq. (23) is plotted in the lower partial illustrations of the Bed Behavior Diagrams. At low filling degrees critical conditions may exist for the occurrence of undesired slipping motion, depending on wall roughness and flow properties of the bed materials.

The ranges and limits of all other types of motion are represented in the upper partial illustrations of the Bed Behavior Diagrams. As is clear from the graphs, the motion behavior of the bed materials studied is similar in principle. Only the transitions of slumping–rolling and rolling–cascading differ in the Froude number by one to two decimal powers and shift to lower Froude numbers with increasing flowability of the solids. An indication of the flowability of granular solids is, among others, the dynamic angle of repose: the smaller the dynamic angle of repose, the more free-flowing the bed. For example sand ($d = 0.5 \text{ mm}$; $\Theta = 32.5^\circ$) possesses the highest flowability

of the bed materials studied. As the flowability of the bed increases, the slumping range diminishes until, when the material is free-flowing, it finally disappears (Fig. 16). As observations confirm [6], slipping motion can turn directly into rolling in this case. Simultaneously, the ranges for rolling and cascading then increase, as a result of which the conditions for heat and mass transfer in the rotary kiln improve. For comparison with measurements, Figs. 14–16 contain experimental values of the slumping–rolling transition from Henein et al. [6]. With the exception of limestone, the predicted boundaries are in agreement with the data. The reason for the differences in case of limestone possibly lies in measuring errors when determining the flow properties or in the simplifications of the mathematical model used.

The transition from cascading motion to cataracting motion, with lower filling degrees ($f < 0.15$), is to be expected only at very high rotational speeds ($Fr > 1$), in

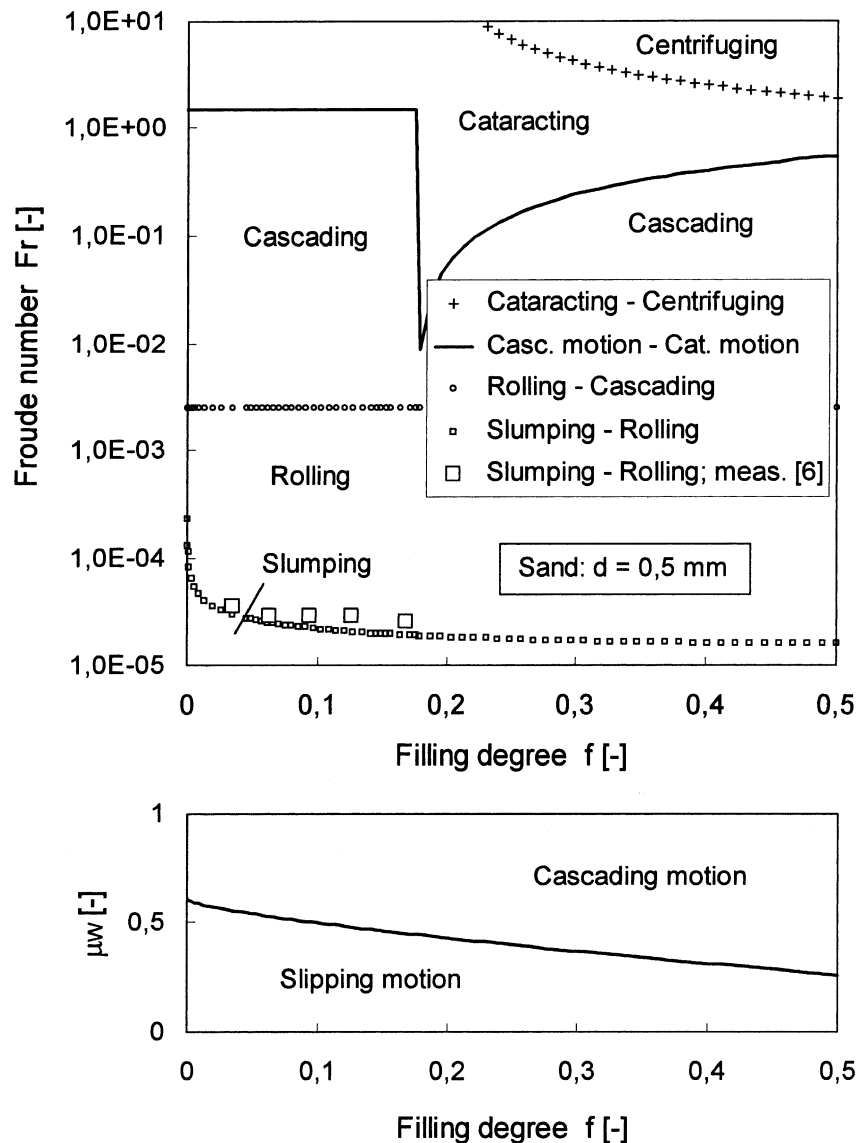


Fig. 16. Bed Behavior Diagram of sand.

accordance with calculations from Eq. (45). For lack of data, a wall friction coefficient of $\mu_w = 0.675$ given for sand in a rotary kiln with sintering zone roughness [25] is applied for all bed materials in the diagrams. At medium filling degrees around $\Theta + \varepsilon = 90^\circ$, a discontinuity given by the model occurs, which in this manner will only happen in the motion of a single particle. However, for the operation of ball mills, only the upper range of the filling degree of $f > 0.2$ is relevant [27], in which the cataracting motion can already be obtained with Froude numbers of $Fr < 1$. These practical experiences correspond well with predictions from Eq. (48). As experiments demonstrate [24], when filling degrees are low, the transition from cataracting to centrifuging cannot be achieved or can only be achieved with extremely high rotational speeds. This

limit approaches the Froude number $Fr = 1$ as the filling degree increases.

6. Summary

This study presents an overview of the forms and the transition behavior of the transverse motion of free-flowing bed materials in unbaffled rotating cylinders. Simplified mathematical models have been developed to describe the transition behavior between the individual forms of motion. Only the transition of rolling–cascading is ascertained through an empirical criterion, for which no suitable mathematical model is known up to now. Criterion equations were derived from the models in the form of critical

wall friction coefficients and critical Froude numbers as a function of the filling degree. Using these criterion equations, the limits between the forms of movement were estimated. Besides the Froude number and the filling degree, the positions of the boundaries are affected to a considerable extent by the flow properties of the bed materials used. It was shown that slipping motion is influenced primarily by the filling degree and the bed/wall friction coefficient, while, for slow rotation, the effect of the Froude number is negligible. Cascading motion and cataracting motion are mainly determined by the Froude number, filling degree, and the bed material properties.

The criterion equations worked out were verified predominantly by comparing them with measurements of experimental rotating cylinders as well as published results from industrial practice. Therefore, data drawn from the literature as well as from this author's own experiments were used. This comparison exhibits an acceptable correspondence between calculation and values from experiments and industrial practice. This study yielded the following scaling criteria for the transverse motion behavior: Froude number, filling degree, coefficient of wall friction, internal frictional coefficient, dynamic and static angle of repose as well as the ratio of particle to rotating cylinder diameter. The parameters of length and inclination of the cylinder, which essentially determine the axial transport, have no influence on the transverse motion, so that generally experimental studies on transverse motion behavior can be conducted batchwise in a horizontal rotating drum.

The transverse motion behavior of bed materials in rotating cylinders postulated above can be conveniently represented in a Bed Behavior Diagram. This Diagram plots the wall friction coefficient and the Froude number against the filling degree and represents the ranges of the individual forms of motion and their limits. It provides the rotary kiln user the possibility to ascertain the motion behavior of the bed material used and, as a result, to influence it by selecting the parameters of rotational speed and filling degree. Continuing studies are necessary to experimentally test the criteria worked out and improve the mathematical models developed.

Nomenclature

A	area (m^2)
d	mean particle diameter; ball diameter (m)
D	diameter of cylinder (m)
Du	dimensionless solids mass flow rate
f	filling degree, defined as fraction of cylinder cross-section filled by solids
f_F	frictional force per unit area (N m^{-2})
F	force (N)
F	force per unit length (N m^{-1})
F	cumulative filling degree
Fr	Froude number
g	gravitational acceleration (m s^{-2})
h	depth of solid bed at mid-chord (m)

h_1, h_2	lengths (m)
l_1, l_2	lever lengths (m)
L	length of cylinder (m)
m	mass (kg)
$\dot{m}$	mass flow rate (kg s^{-1})
M	mass per unit length (kg m^{-1})
M	moment per unit length (N)
n	rotational speed (min^{-1})
r	radius (m)
r_0	distance cylinder axis—bed surface (m)
$r_{0,x}$	distance cylinder axis—point B (line OB), see Fig. 1 (m)
r_S	radius of the center of gravity of the bed (centroid) (m)
R	radius of cylinder (m)
R_i	inner radius of ball ring (m)
s	chord of the solid bed (m)
s^*	medium distance traveled by particles in a slump (line S_1S_2 , see Fig. 3) (m)
t	time (s)
t_1, t_2	lifting time and slumping time, respectively (s)
T	mean residence time (s)
v	particle velocity (m s^{-1})
x	distance traveled by particles (m)
α	angle of repose
$\Delta\alpha$	shear wedge angle
γ_0	measured shear wedge angle [6]
δ	angle of deflection of the center of gravity of the bed
ε	filling angle, defined as half the angle of sector occupied with solids
ζ	angle coordinate
η	angle of inclination of particle trajectory
Θ	dynamic angle of repose
Θ_0	static angle of repose
κ	angle coordinate; angle of deflection of a single particle
μ	friction coefficient
μ_i	coefficient of internal friction
μ_w	friction coefficients particle/wall and bed/wall, respectively
ρ_b	bulk density (kg m^{-3})
φ_i	angle of internal friction
φ_w	angle of friction of particle/wall and bed/wall, respectively
ω	angular rotation speed (s^{-1})

Subscripts

c	critical
C	centrifugal
F	frictional
g, G	gravitational
i	internal
I	inertial
L	lower

m	medium
N	normal
P	particle
S	solid(s); center of gravity
U	upper
W	wall
x	transversal

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